DECISION TREES

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Introduction to Decision Trees

Definition

Decision trees are a supervised machine learning algorithm used for both classification and regression tasks.

Decision Trees model decisions and their possible consequences in a tree-like structure, where each internal node represents a decision based on a feature, each branch represents an outcome of that decision, and each leaf node represents a final prediction

Multivariate Decision Trees split on a combination of features

Key Characteristics

- Decision Trees split data based on rules or questions
- Decision Trees are used in both classification and regression
- Decision Trees are easy to interpret and visualize

Introduction to Decision Trees

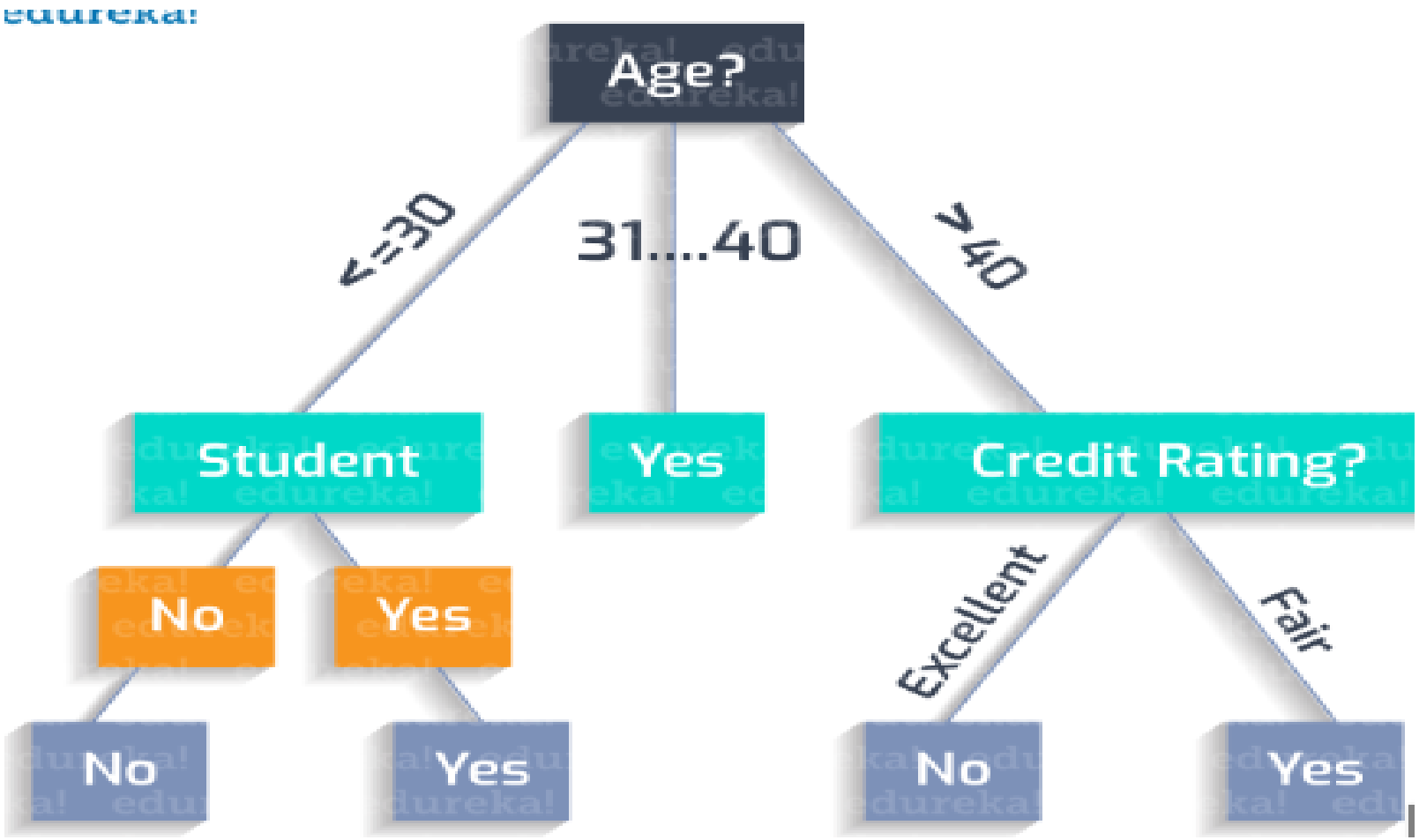
When To use Decision Trees

Decision Trees are used you need a model that is easy to understand.

Decision Trees work well when the data has a combination of both categorical and continous data.

Decison Trees also work best when one expects non-linear relationships. If the relationship between variables is not a straight line, trees are helpful because the decision trees can capture complex patterns and split the data in flexible ways.

Decision Trees Sample



History of Decision Trees

- **1960s:** Early origins of Decision Trees used in operations research and decision analysis for management decisions.
- **1980s:** Introduction of of formal machine learning decision tree algorithms:ID3 (1986) by Ross Quinlan and CART (1984) by Breiman.
- **1990s:** Improvement of ID3 where C4.5 was introduced by Quinlan. C4.4 handles continuous attributes, pruning, and missing values.
- **2000s:**: Decision trees became widely used in machine learning, integrated into ensemble methods like Random Forests and Gradient Boosting.
- **Present:** Continues to be a popular and effective algorithm. The algorithm has been used in classification, regression and also integrated with other machine learning models. The model has been popular with datasets due to its high interpretability and high accuracy.

Definition of Terms

1. Root Node

This is the starting point. It represents the entire dataset.

2. Branches

These are the lines connecting the nodes showing the flow from one decision to another.

3. Internal Nodes

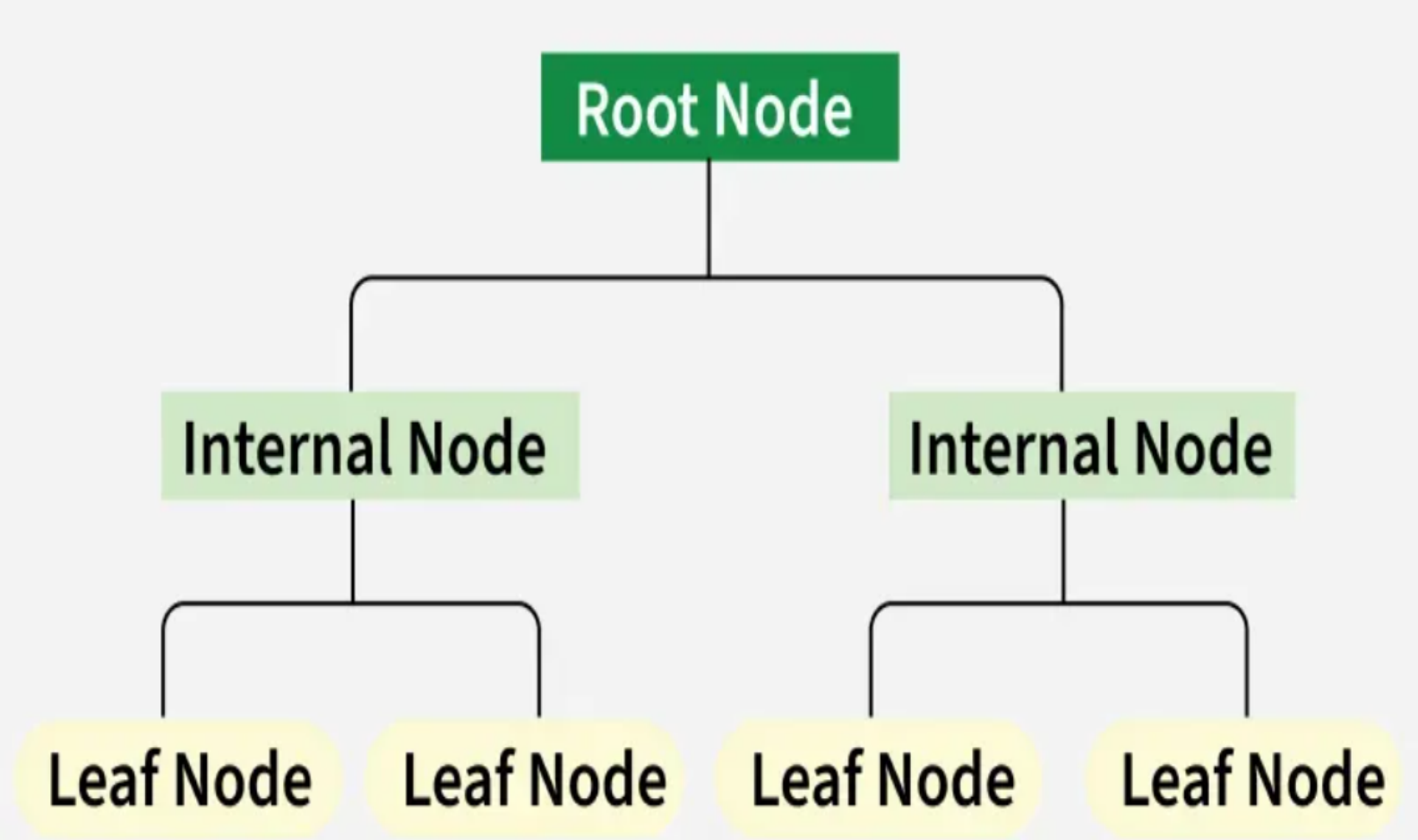
Points where decisions are made based on data features.

4. Leaf Nodes

End points of the tree showing where the final decision or prediction is made.

Decision Tree Visualization

Decision Trees Concept



Types of Decision Trees

1. Iterative DIchotomiser (ID3)

ID3 is a type of decision tree that measures the degree of data mixing at a node using entropy. ID3 uses:

- Entrophy : measures amount of uncertainity or disorder in the data.
- Information Gain : Tells how much the data entrophy reduces after splitting the data using a feature.

When to use Id3

- When the dataset has categorical features (like color, type, or yes/no attributes).
- When the target variable has discrete classes (distinct categories like "Yes" / "No" or "Cat" / "Dog").

How ID3 Works: Step by Step

Core Process:

1. **Data Preparation**
   - Collect and clean the dataset. The data should have labels (Y) and input features (X).
2. **Calculate Entropy and Information Gain**
   - Compute the entropy of the target variable first:

$$H(S) = -\sum_{i=1}^{c} p_i \log_2 p_i$$

   where c is the number of classes and p_i is the proportion of examples in class i.
   - For each feature, calculate the expected entropy and then the information gain:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

   where S_v is the subset of S where feature A has value v.

4. **Split the Dataset**
   - Divide the dataset into subsets based on the selected feature's values.
   - Each subset corresponds to a branch in the decision tree.
5. **Recursive Tree Building**
   - Repeat steps 2-4 for each subset.
6. **Stopping Condition**
   - Stop when all subsets are pure, no features remain, or subsets are empty.
   - Create a decision tree that classifies new examples by following paths from the root to the leaf nodes.

Problem Statement & Dataset

Goal: Predict Field Activity based on weather conditions

Season	Temp	Humidity	Wind	Field Activity
sunny	mild	high	no	no
sunny	warm	high	no	no
sunny	warm	high	yes	no
rainy	mild	high	yes	no
rainy	fresh	standard	yes	no
cloudy	warm	high	no	Yes
rainy	mild	high	no	Yes
cloudy	warm	standard	no	Yes
rainy	mild	standard	no	Yes
sunny	fresh	standard	no	Yes
rainy	fresh	standard	no	Yes
cloudy	mild	high	yes	Yes
sunny	mild	standard	yes	Yes
cloudy	fresh	standard	yes	Yes

- Total examples: 14, Yes: 9, No: 5
- Initial Entropy: $H(S) = 0.9403$ bits

Information Gain Calculations

Calculate IG for all features:

- **Season:**
  - sunny: 2Y,3N ($H = 0.971$), rainy: 4Y,2N ($H = 0.918$), cloudy: 3Y,0N ($H = 0$)
  - Weighted: $(5/14) \times 0.971 + (6/14) \times 0.918 + (3/14) \times 0 = 0.7402$
  - $\mathbf{IG = 0.9403 - 0.7402 = 0.2001}$
- **Humidity:**
  - high: 4Y,4N ($H = 1.000$), standard: 5Y,1N ($H = 0.650$)
  - Weighted: $(8/14) \times 1.000 + (6/14) \times 0.650 = 0.8500$
  - $IG = 0.9403 - 0.8500 = 0.0903$
- **Wind:** $IG = 0.0483$
- **Temperature:** $IG = 0.0499$

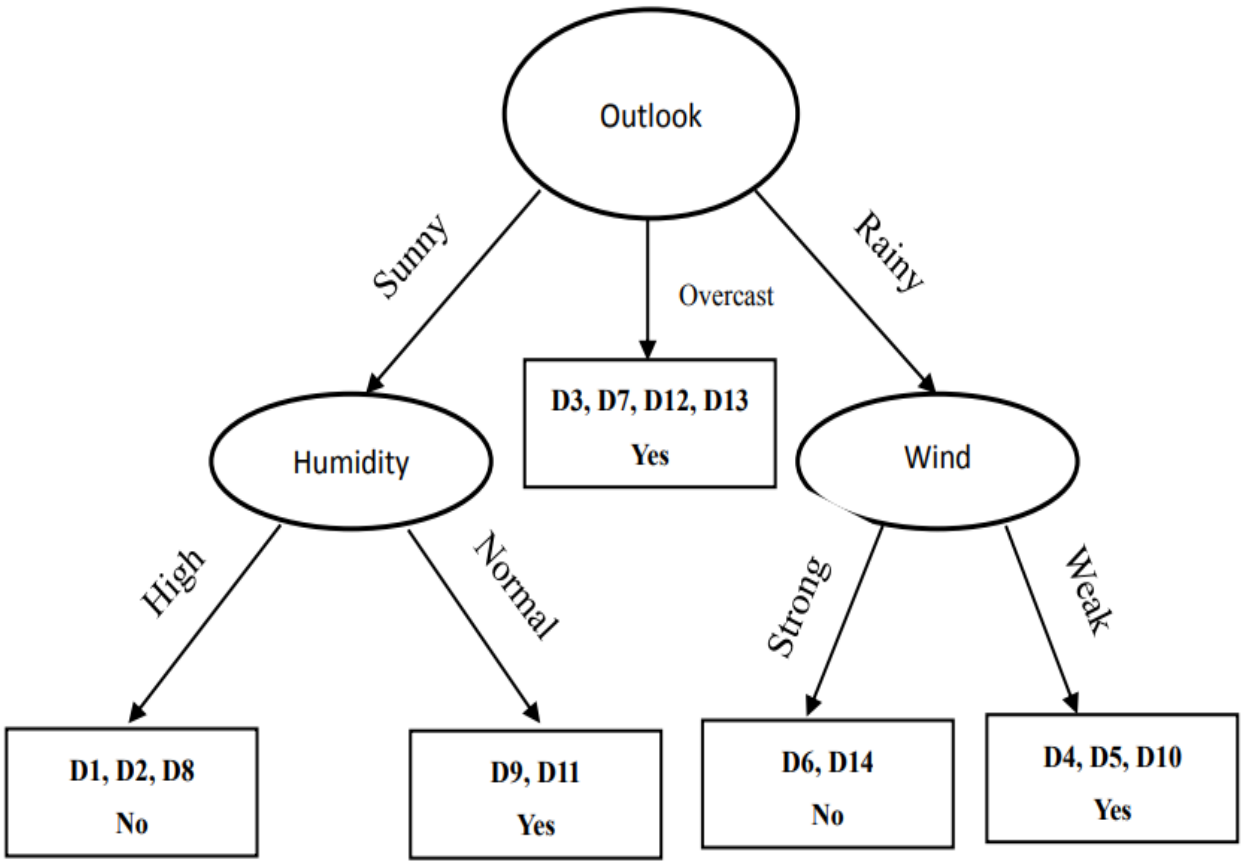
Root Node Selection

Feature	Information Gain
Season	0.2001
Humidity	0.0903
Temperature	0.0499
Wind	0.0483

Root Node

Season has highest Information Gain → Root Node. Repeat steps the process of finding entrophy and information gain for each subset.Stop when all subsets are pure, no features remain, or subsets are empty.Create a decision tree that classifies new examples by following paths from the root to the leaf nodes.

Decision Tree



Types of Decision Trees

2. C4.5

C45 is a decision tree algorithim which is an improvement of ID3. Its main purpose is to classify data by creating a decision tree based on a set of attributes.

- Can handle both continuous and discrete attributes
- Handles missing values
- C4.5 can perform tree pruning to avoid overfitting
- C4.5 Uses Gain Ratio for Splitting

C4.5 builds a decision tree by choosing splits that best separate the data and continues splitting until the subsets become pure or cannot be divided further.

C4.5 is the same as ID3 and its more powerful and works best with messy data.

How C4.5 Works: Step by Step

Core Process:

1. **Data Preparation**
   - Collect and clean the dataset. The data should have labels (Y) and input features (X).
2. **Calculate Entropy and Information Gain**
   - Compute the entropy of the target variable first:

$$H(S) = -\sum_{i=1}^{c} p_i \log_2 p_i$$

where c is the number of classes and p_i is the proportion of examples in class i.

   - For each feature, calculate the expected entropy and then the information gain:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

where S_v is the subset of S where feature A has value v.

4. **Select the Best Attribute to Split**
   - Partition the dataset into subsets based on the chosen attribute's values. If the attribute is continuous, C4.5 finds the best threshold to split the data into two groups where one group is less than or equal to the threshold and the other group is greater than the threshold)
5. **Recursive Tree Building**
   - Repeat steps 2-4 for each subset.
   - The recursion goes on like a loop until :
     - All instances in the subset belong to the same class.
     - There are no features left to split on.
     - The dataset is empty.
6. **Tree Pruning**
   - Remove trees that are not necessary to improve classification accuracy on unseen data.
7. **Evaluation**
   - Evaluate the tree using well-known metrics such as accuracy, precision, recall, F1-Score,
   - Create a decision tree that classifies new examples by following paths

Example

This demonstration uses the C4.5 decision tree algorithm to predict job offer outcomes for students based on academic and personal attributes. The classification goal is to determine whether a student receives a job offer (Yes/No) using GPA, interactiveness, practical knowledge, and communication skills.

GPA	Interact.	Prac. Know.	Comm. Skills	Job Offer
≥ 9	Yes	V. good	Good	Yes
≥ 8	No	Good	Moderate	Yes
≥ 9	No	Average	Poor	No
< 8	No	Average	Good	No
≥ 8	Yes	Good	Moderate	Yes
≥ 9	Yes	Good	Moderate	Yes
< 8	Yes	Good	Poor	No
≥ 9	No	V. good	Good	Yes
≥ 8	Yes	Good	Good	Yes
≥ 8	Yes	Average	Good	Yes

Step 1: Calculate Initial Entropy

Total instances: $N = 10$

Job Offer: Yes = 6, No = 4

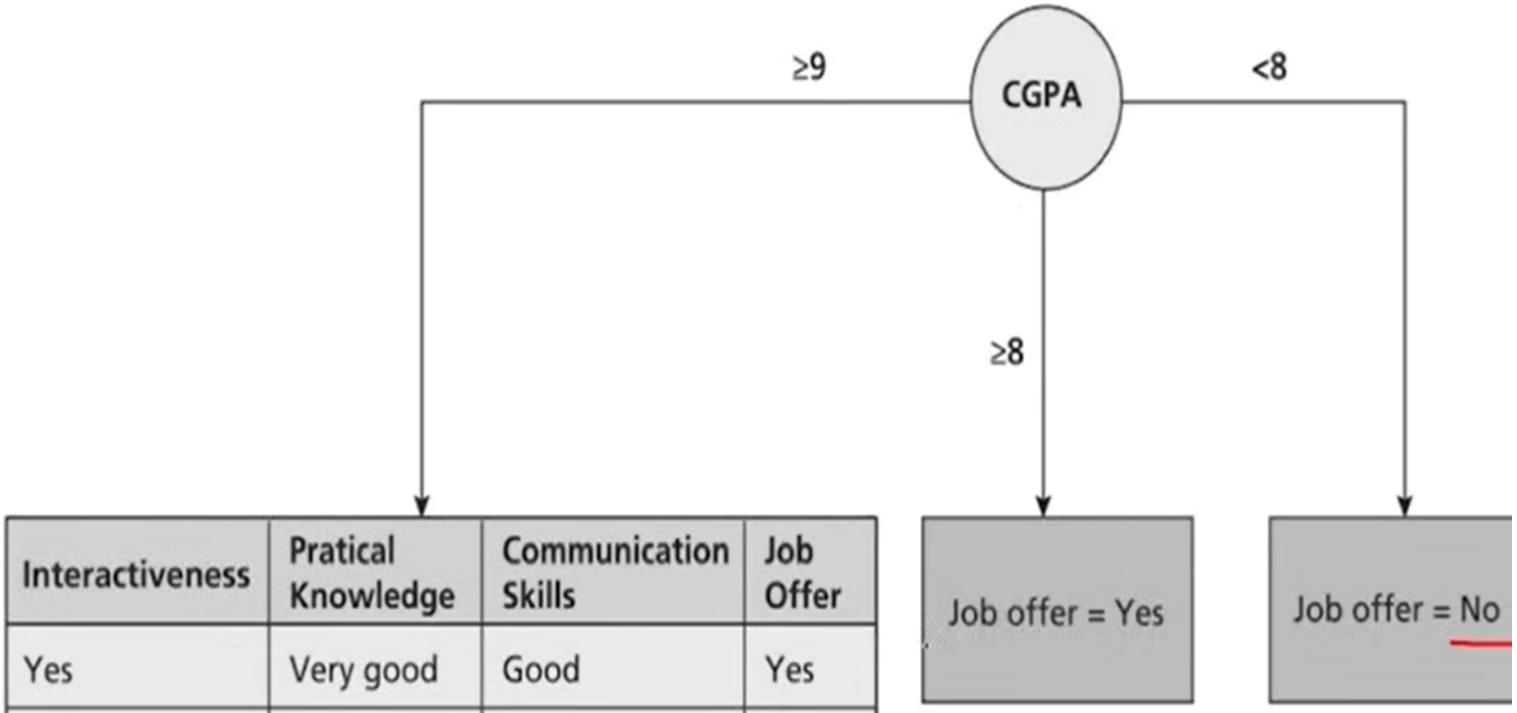
$$\text{Entropy}(T) = -\frac{7}{10}\log_2\left(\frac{7}{10}\right) - \frac{3}{10}\log_2\left(\frac{3}{10}\right)$$
$$= 0.8807$$

Step 2: Calculate Gain Ratio for Each Attribute

Attribute	**Gain**	**Split Info**	**Gain Ratio**
CGPA	0.5564	1.1521	0.3658
Interactiveness	0.0911	0.9704	0.0939
Practical Knowledge	0.2448	1.4853	0.1648
Communication Skills	0.5202	1.4853	0.3502

Decision: CGPA has the highest Gain Ratio (0.3658), so it becomes the root node. The split on CGPA instantly created two pure subsets (CGPA < 8 and CGPA ≥ 8 and < 9). The third subset (CGPA ≥ 9) remains impure, meaning a high CGPA alone is not enough to classify the outcome and more splitting needs to be done there.

Subset Data for CGPA < 9



Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

Step 3: Calculate Entropy for Job Offer again

Total instances: $N = 4$

Job Offer: Yes $= 3$, No $= 1$

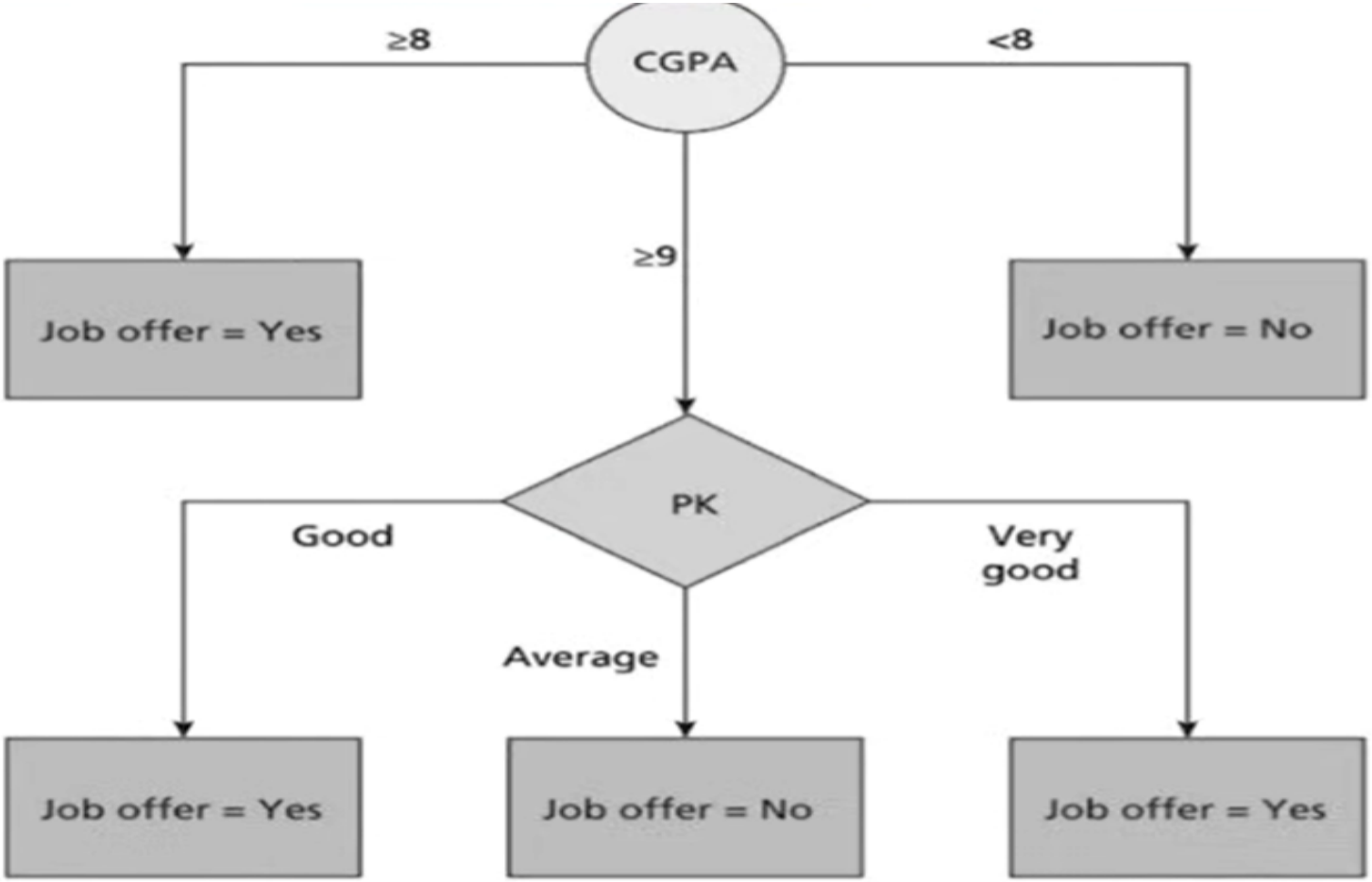
$$\text{Entropy}(T) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$$
$$= 0.8112$$

Step 4: Calculate Gain Ratio for Remaining Attributes

Attribute	**Gain**	**Split Info**	**Gain Ratio**
Practical Knowledge	0.8108	1.5	0.5408
Interactiveness	0.311	1.0	0.3112
Communication Skills	0.8108	1.500	0.5408

Final Decision: Interactiveness and Communication Skills has the highest Gain Ratio (0.5408) and the splits will occur there.

Final Tree Diagram



Types of Decision Trees

3. CART

CART is a powerful decision tree algorithm that can handle both classification and regression.

- CART uses Gini Impurity for classification and variance reduction for regression.
- CART always produces binary trees. Each split divides the data into two branches, even for multiple-category features.
- CART can work for both continous and categorical features.

CART builds binary decision trees by repeatedly splitting the dataset into two groups based on the feature that best separates the classes
CART also uses tree pruning to reduce overfitting.

How CART Works: Step by Step

Core Process:

1. **Data Preparation**
   - Collect and clean the dataset. The data should have labels (Y) and input features (X). Take all the training data as the root node
2. **Selecting the Best Split**
   - Compute the Gini Index For each feature, compute the Gini Index and choose the split that minimizes impurity. Gini(D) = 1 - $\sum_{i=1}^{C} p_i^2$
3. **Feature Selection**
   - Select the feature with the highest information gain.
   - This feature becomes the decision node at the current level of the tree.

4. **Split the Dataset and do Data Partitioning**
   - Divide the dataset into subsets based on the selected feature's values.
   - Each subset corresponds to a branch in the decision tree.
   - Split the dataset into two subsets based on the selected rule Create left node and right node

5. **Recursive Tree Building**
   - Repeat steps 2-4 for each subset.

6. **Stopping Condition and Tree Pruning**
   - Stop when all subsets are pure, no features remain, or subsets are empty.
   - Create a decision tree that classifies new examples by following paths from the root to the leaf nodes.
   - Apply pruning to remove branches that do not improve predictive accuracy, reducing overfitting.

Advantages of Decision Trees

Key Advantages

- **E**asy to understand and interpret
- **C**an handle both continous and categorical data
- **R**equires little data preprocessing steps. For instance, there is no need for standardization or min-max scaling.
- **C**an capture non-linear patterns between features and target.
- **P**rovides insight into which features are most influential in predictions.

Practical Benefits

- Fast training on medium-sized datasets
- Decision trees has Built-in feature importance scores
- Decision trees can model non-linear relationships

Disadvantages of Decision Trees

Key Challenges

- **D**ecision trees can create very complex trees that fit the training data too closely, reducing generalization to new data hence leading to overfitting.
- **T**here is instability in that Small changes in the data can lead to a completely different tree structure
- **T**rees can be biased toward features with more levels or continuous variables.

Practical Limitations

- Single decision trees are often less accurate than ensemble methods like Random Forests or Gradient Boosting.
- Complex Trees Are Hard to Interpret
- Sensitive to Noisy Data

Limitations of Decision Trees

Main Limitations

- **Instability Issues:** Small data changes may lead to completely different trees
- **Inherent Biases:** Biased toward attributes with more levels. Struggles with diagonal decision boundaries
- to overfit without proper constrain
- **P**oor at learning linear relationships and can become very deep with many features.

Despite these limitations, Decision Trees remains widely used for its robustn

Evaluation Metrics in Decision Trees

Classification Metrics

- **Accuracy:** Proportion of correctly classified instances. Formula: Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** Proportion of positive predictions that are correct. Formula: Precision $= \frac{TP}{TP+FP}$
- **Recall:** Proportion of actual positives correctly identified. Formula: Recall $= \frac{TP}{TP+FN}$
- **F1-score:** Harmonic mean of Precision and Recall. Formula: $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}}$

Regression Metrics

- **Mean Squared Error (MSE):** Average squared difference between actual and predicted values. Formula: MSE $= \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
- **Mean Absolute Error (MAE):** Average absolute difference between actual and predicted values. Formula: MAE $= \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
- **R^2:** Proportion of variance in the dependent variable explained by the model.

Decision Tree Algorithms: Summary

Algorithm	Splitting Criterion	Key Features	Best Use Cases
ID3	Information Gain	• Simple categorical data • Prone to overfitting	Small datasets Categorical features
C4.5	Gain Ratio	• Handles numeric data • Supports missing value • Built-in pruning	Mixed data types
CART	Gini Index / Variance	• Binary splits only Classification and Regression • Robust pruning	Binary decisions Regression tasks

When to use Each Algorithim

- **ID3** - Simple categorical data
- **C4.5** - Mixed data types
- **CART** - Regression or binary splits

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Conclusion

Thank You!

Questions?